

① Find avg of 3 numbers.

1. input $\rightarrow$ 3 integers

2. process $\rightarrow$ logic / condition $\left\{ \begin{array}{l} \text{Sum} \\ \text{sum/n} \end{array} \right.$

3. output $\rightarrow$ avg

Algorithm :

step ①: start

step 2: Take 3 numbers & store in sum

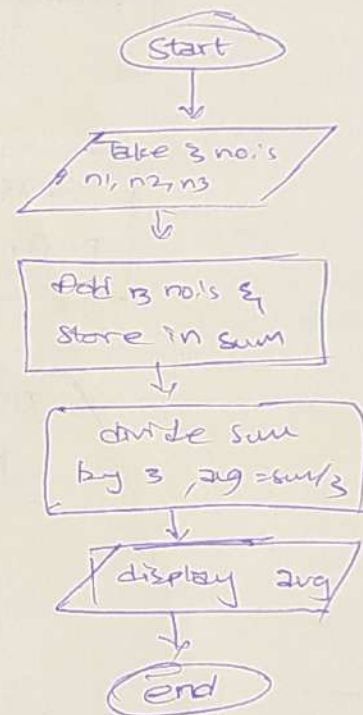
step 3: Add 3 no.s & store in sum

step 4: divide sum by 3

step 5: display avg

step 6: end

Flowchart :



2. Find the profit percentage
by using selling & cost price.

① input $\rightarrow$ SP, CP

② process $\rightarrow$ $\left\{ \begin{array}{l} \text{profit} = \text{SP} - \text{CP} \\ \text{PP} = \frac{\text{P}}{\text{CP}} \times 100 \end{array} \right.$

③ output $\rightarrow$ Profit %

Algorithm :

Step 1: start

Step 2: Take inputs SP, CP

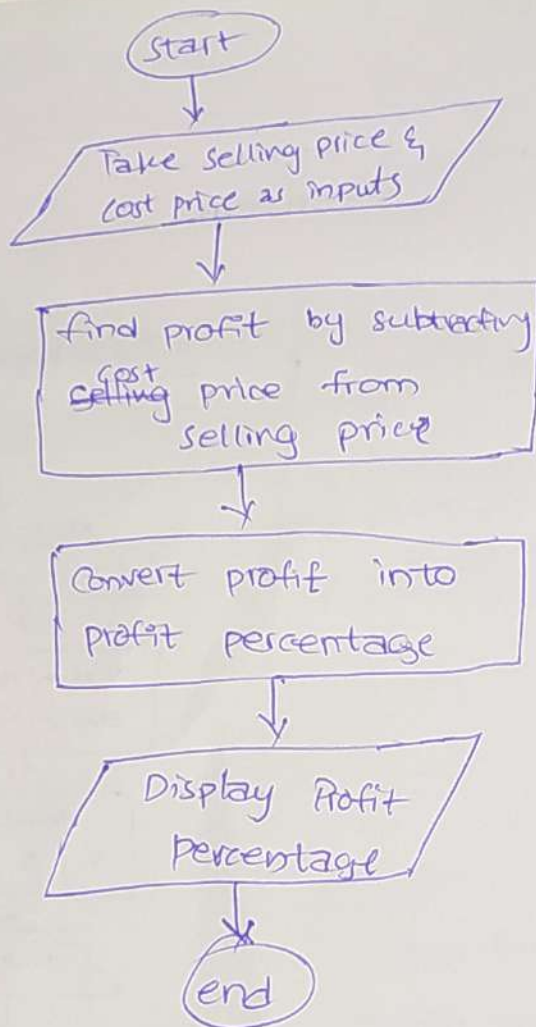
Step 3: find profit by subtracting CP from SP

Step 4: Convert profit into profit percentage by dividing profit by cost price then multiply with 100 ($\text{PP} = \frac{\text{P}}{\text{CP}} \times 100$)

Step 5: Display profit percentage.

Step 6: End.

flowchart :



CP = 100
SP = 120

$$\frac{20}{100} \times 100 = 20\%$$

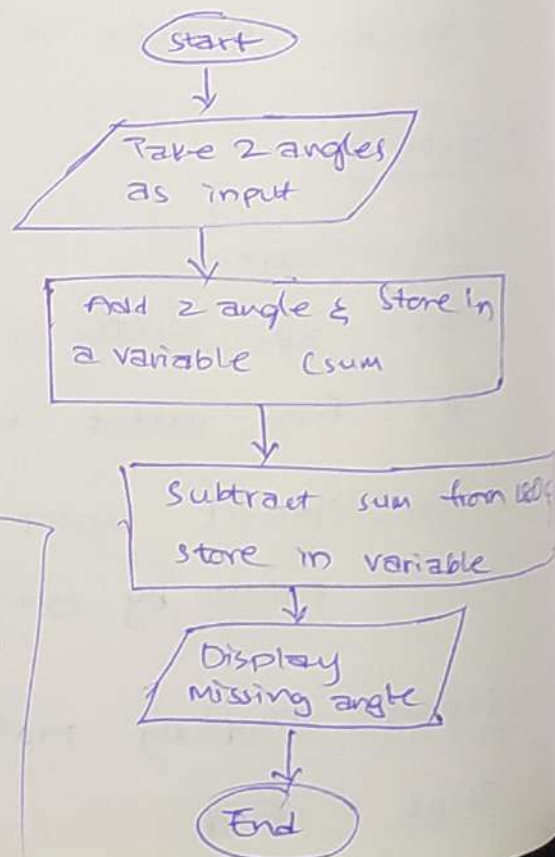
3. Find the missing angle in triangle by taking 2 angles

- ① input → 2 angles of triangle
- ② process → Add 2 angles, store in sum
Subtract sum from 180
- ③ output → missing angle.

Algorithm :

- Step 1 : start
- Step 2 : Take two sides of Δ as i/p's
- Step 3 : Add 2 angle & store in a variable (sum)
- Step 4 : Subtract sum from 180, store in variable (missing angle)
- Step 5 : Display missing angle
- Step 6 : End

Flowchart :



4. Find the last digit of given number.

→ input : Take a number

→ process : — Divide no. with 10 & store it
consider remainder

→ output : last digit

algorithm :

step 1 : start

step 2 : Divide Take a number

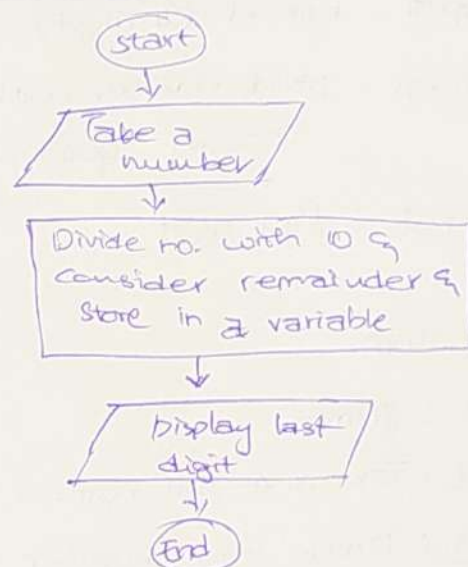
step 3 : Divide the number with 10 &
consider remainder.

step 4 : store remainder in a variable

step 5 : Display last digit i.e, remainder

step 6 : End.

flowchart :



5. Remove last digit from given number.

→ input : Take a number

→ process : divide number with 10 & consider quotient

→ output : Number without last digit

algorithm :

step 1 : start

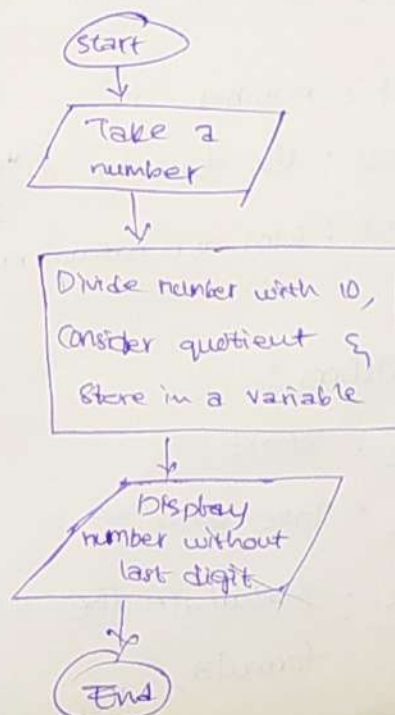
step 2 : Take a number

step 3 : Divide number with 10 &
consider quotient & store in a variable

step 4 : Display number without last digit

step 5 : End

flowchart :



6. Find the first digit of 4 digit number
7. Find sum of first 'n' natural numbers. (without using loops)
8. Find avg of first 'n' natural numbers.

6. Find the first digit of 4 digit number.

input : number (4 digit)

process : Divide number with 1000 &
Consider quotient

output : first digit

Algorithm :

Step 1 : Start

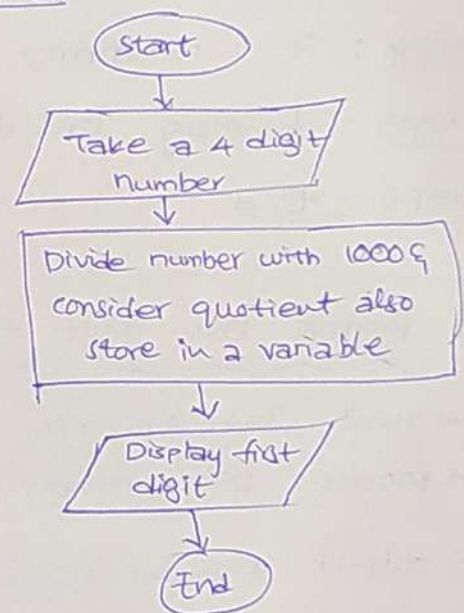
Step 2 : Take a 4 digit number

Step 3 : Divide the number with 1000 &
Consider quotient & store in a variable

Step 4 : Display the first digit

Step 5 : End

flowchart :



7. Find sum of first 'n' natural numbers.

→ input : number 'n'

→ process : Use formula $\frac{n*(n+1)}{2}$

→ output : sum of 'n' natural numbers

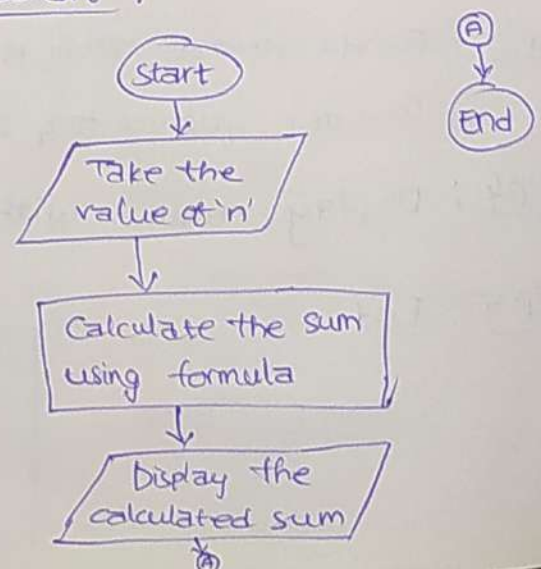
Algorithm :

Step 1 : Start

Step 2 : Take value of 'n'

Step 3 : Calculate the sum using formula,
 $Sum = \frac{n(n+1)}{2}$

flowchart :



step : Display the calculated sum.

step5 : End

8. find avg of first 'n' natural numbers

⇒ input : number 'n'

⇒ process : $\left\{ \begin{array}{l} \text{find sum of 'n' natural no's using } \frac{n(n+1)}{2} \\ \text{then divide sum with n i.e. sum/n} \end{array} \right.$ (or) directly $\frac{\frac{n(n+1)}{2}}{n} = \frac{n+1}{2}$ (avg)

⇒ output : avg of first 'n' natural no's

Algorithm :

step1 : Start

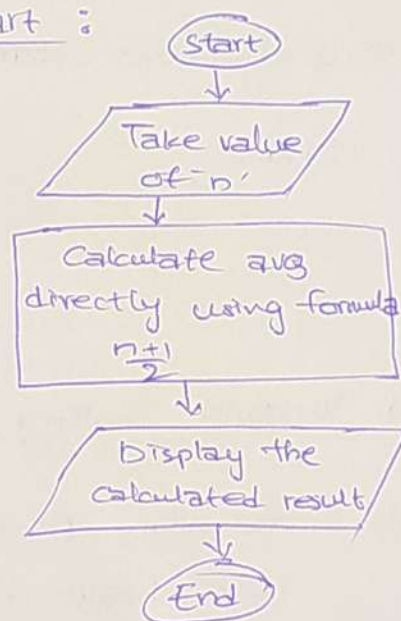
step2 : Take value of 'n'

step3 : Use formula, $\frac{n+1}{2}$ & store in a variable

step4 : Display the calculated value

step5 : End

flowchart :



04/08/26

9. find gross salary

① basic salary = 20000

② bonus = 20%

③ incentive = 5%

⇒ input : basic salary, bonus, incentive

⇒ process :⇒ logic $\left\{ \begin{array}{l} \text{bonus} \rightarrow 20\% = \frac{20}{100} \times 20000 = 4000 \\ \text{incentive} \rightarrow 5\% = \frac{5}{100} \times 20000 = 1000 \\ \text{gross} = \text{b.s} + \text{bonus} + \text{incentive} = 25000 \end{array} \right.$

⇒ output : Gross salary.

Algorithm :

step1 : Start

step2 : Take basic salary, bonus^{percentage} and incentives^{percentage} as input.

Step 3: Convert bonus % & incentive % into actual amount by using,

$$\text{bonus} = \frac{\text{bonus \%}}{100} \times \text{b.s}, \text{ incentive} = \frac{\text{incentive \%}}{100} \times \text{b.s}$$

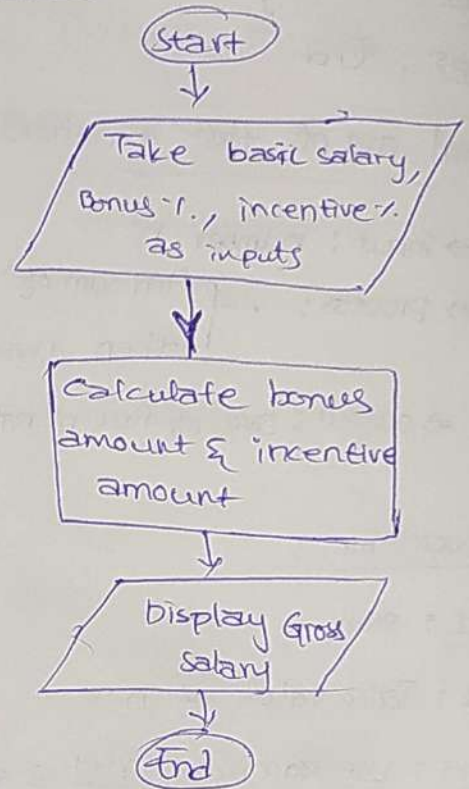
Step 4: Calculate gross salary using,

$$\text{gross salary} = \text{Basic salary} + \text{Bonus Amount} + \text{Incentive Amount}$$

Step 5: Display Gross salary.

Step 6: End

flowchart:



10. Find in-hand salary.

→ input: basic salary, bonus %, incentive %, PF %, health insurance %

→ process: find bonus, incentive, PF & health insurance amount.
Add b.s, bonus, incentive, we get gross salary & subtract PF & health insurance from g.s

→ output: inhand salary.

Algorithm:

Step 1: start

Step 2: Take basic salary, bonus %, incentive %, PF %, health insurance %

$$\begin{array}{l|l} \text{B.S} = 20000 & \text{PF \%} = 2 \\ \text{bonus \%} = 20 & \text{health insurance \%} = 1 \\ \text{incentive \%} = 5 & \end{array}$$

$$\text{b.Amount} = \frac{20}{100} \times 20000 = 4000$$

$$\text{ince. amt} = \frac{5}{100} \times 20000 = 1000$$

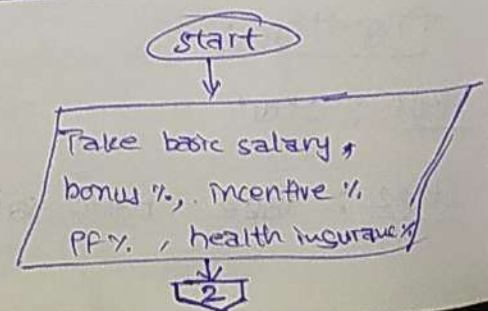
$$\text{PF amt} = \frac{2}{100} \times 20000 = 400$$

$$\text{H.I amt} = \frac{1}{100} \times 20000 = 200$$

$$\text{g.s} = 20000 + 4000 + 1000 = 25000$$

$$\begin{aligned} \text{inhand salary} &= \text{g.s} - \text{PF amt} - \text{H.I amt} \\ &= 25000 - 400 - 200 \\ &= 24400.0 \end{aligned}$$

flowchart:



step3: Calculate the bonus amount, incentive

amount, gross salary, PF amount, health insurance amount, using formulas, $\frac{\%}{100} \times \text{basic salary}$

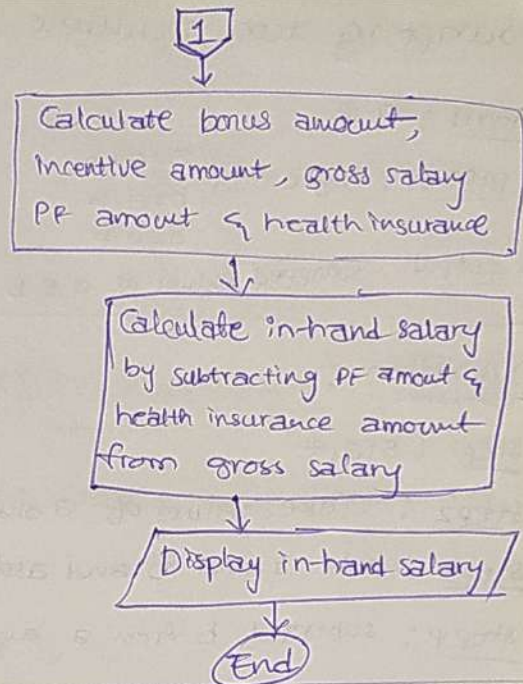
step4: calculate in-hand salary,

$$GS = \text{basic salary} + \text{bonus amount} + \text{incentive amount}$$

$$\text{in-hand} = GS - \text{PF amount} - \text{health insur. and}$$

step5: Display in-hand salary.

step6: End



11. Swap two numbers using three variables

i/p $a=10$ wrong code

$b=20$

o/p:

$a=20$
 $b=10$

$a=b$
 $b=a$

$a \rightarrow 10$

$b \rightarrow 20$

$10/20$

$20/20$

→ input: a, b

→ process: swap $\begin{cases} c=a \\ a=b \\ b=c \end{cases}$

→ output: a & b after swapping

Background

$a=10$

10

$b=20$

20

#logic

$c=a$

10

$a=b$

20

$b=c$

10

$a=20$

$b=10$

Algorithm:

step1: Start

step2: Take the value of a, b as inputs,

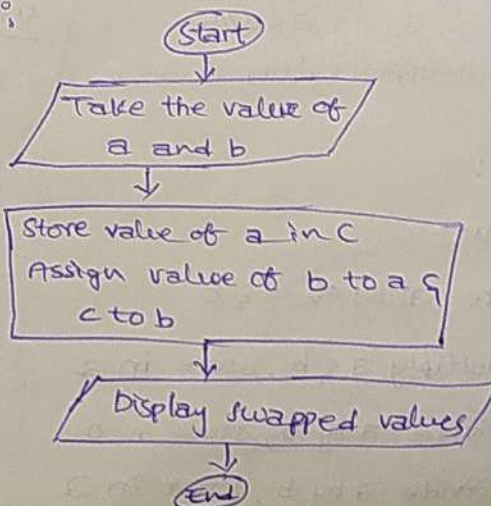
step3: store value of a in c .

step4: Assign value of b to a and c to b .

step5: Display the swapped values of a & b

step6: End

Flowchart:



Swapping two numbers using two variables

→ input : a, b

→ process : logic $\rightarrow \begin{cases} a = a + b \\ b = a - b \\ a = a - b \end{cases}$

→ output : swapped values of a & b

Algorithm :

Step1 : Start

Step2 : Take values of a and b

Step3 : Add a and b and assign to a

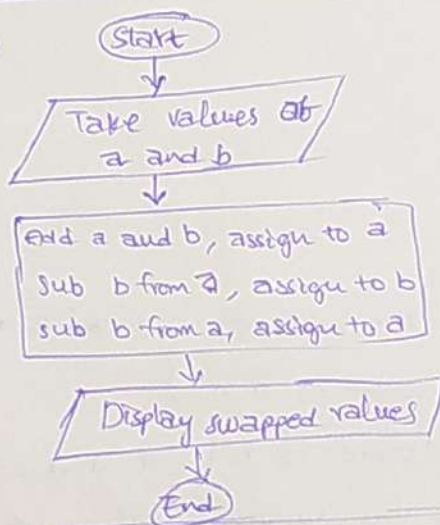
Step4 : subtract b from a and assign to b

Step5 : subtract b from a and assign to a

Step6 : Display swapped values

Step7 : End

flowchart :



Input : a, b

Process : logic $\rightarrow \begin{cases} a = a * b \\ b = a // b \\ a = a // b \end{cases}$

output : swapped values of a & b

algorithm :

Step1 : start

Step2 : Take values of a & b

Step3 : multiply a & b, store in a

Step4 : Divide a by b, store in b

Step5 : Divide a by b, store in a

Background :

i/p : a = 12, b = 8

$$a = a \times b = 12 \times 8 = 96$$

$$a \rightarrow \boxed{96}$$

$$b = a // b = 96 // 8 = 12$$

$$b \rightarrow \boxed{12}$$

$$a = a // b = 96 // 12 = 8$$

$$a \rightarrow \boxed{8}$$

o/p : a = 8
b = 12

Background

i/p

$$a = 10$$

$$b = 20$$

$$\Rightarrow a = a + b$$

$$a = 10 + 20$$

$$a = 30$$

$$\boxed{30}$$

a

$$\Rightarrow b = a - b$$

$$b = 30 - 20$$

$$b = 10$$

$$\boxed{10}$$

b

$$\Rightarrow a = a - b$$

$$a = 30 - 10$$

$$a = 20$$

$$\boxed{20}$$

a

$$\text{o/p: } a = 20$$

$$b = 10$$

Step6 : Display swapped values of a & b

Step7 : End

flowchart :

